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EEM 561 Machine Vision

# Lecture 7

## Corners

Spring 2025

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# Keypoint extraction: Corners

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# Why extract keypoints?

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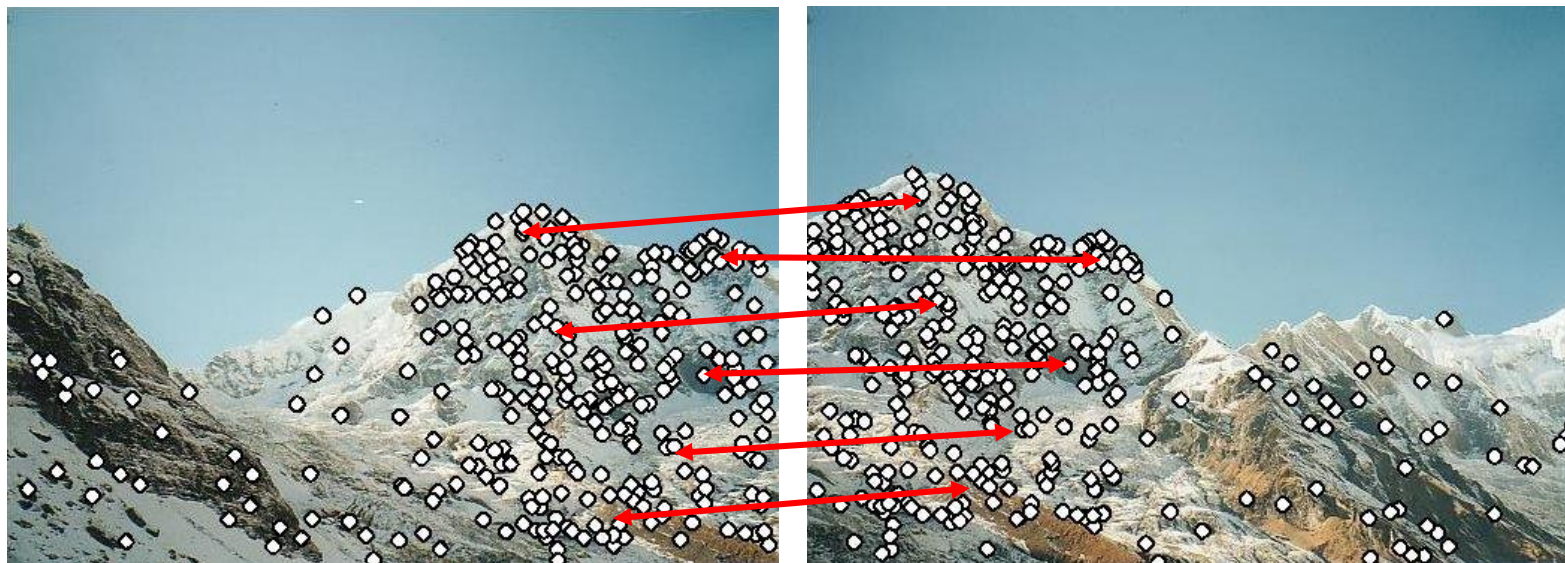
- Motivation: panorama stitching
  - We have two images – how do we combine them?



# Why extract keypoints?

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- Motivation: panorama stitching
  - We have two images – how do we combine them?



Step 1: extract keypoints

Step 2: match keypoint features



# Why extract keypoints?

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- Motivation: panorama stitching
  - We have two images – how do we combine them?



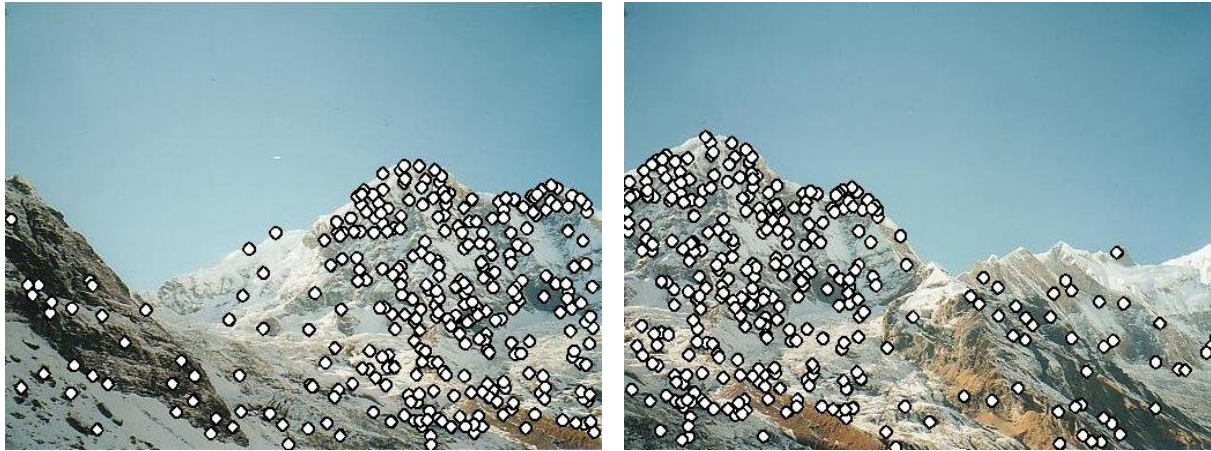
Step 1: extract keypoints

Step 2: match keypoint features

Step 3: align images

# Characteristics of good keypoints

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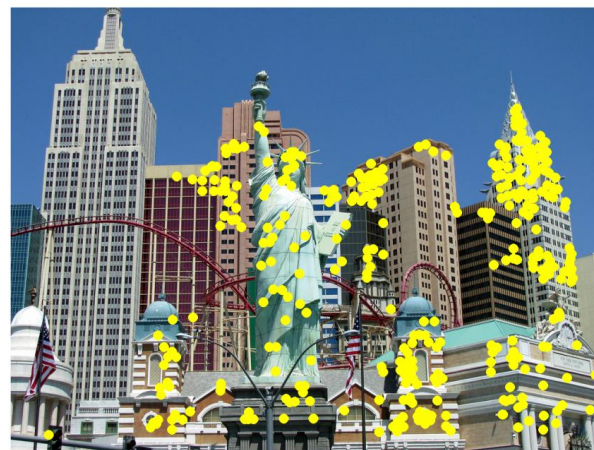
- **Compactness and efficiency**
  - Many fewer keypoints than image pixels
- **Saliency**
  - Each keypoint is distinctive
- **Locality**
  - A keypoint occupies a relatively small area of the image; robust to clutter and occlusion
- **Repeatability**
  - The same keypoint can be found in several images despite geometric and photometric transformations

# Applications

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Keypoints are used for:

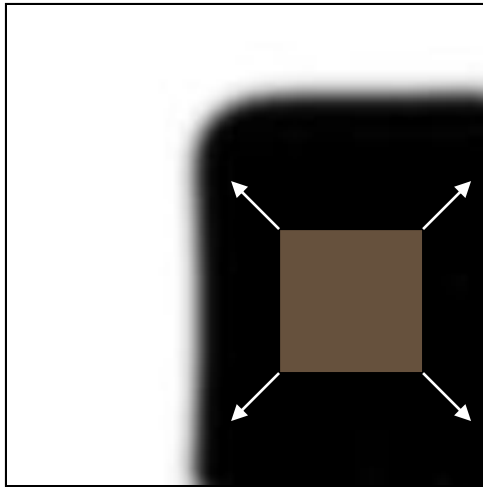
- Image alignment
- 3D reconstruction
- Motion tracking
- Robot navigation
- Indexing and database retrieval
- ~~Object recognition~~



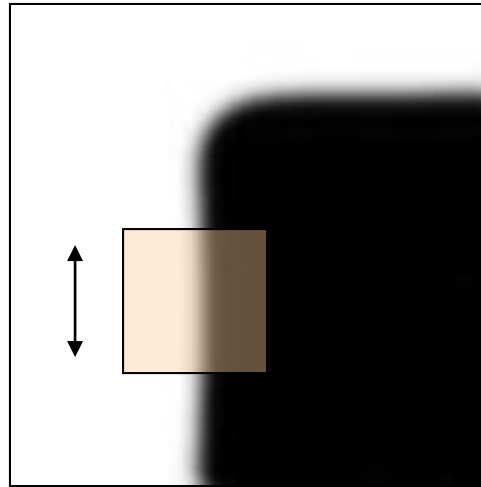
# Corner Detection: Basic Idea

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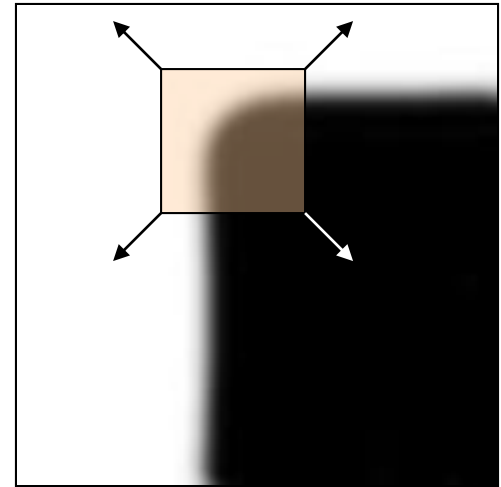
- We should easily recognize the point by looking through a small window
- Shifting a window in *any direction* should give *a large change* in intensity



“flat” region:  
no change in  
all directions



“edge”:  
no change  
along the edge  
direction



“corner”:  
significant  
change in all  
directions

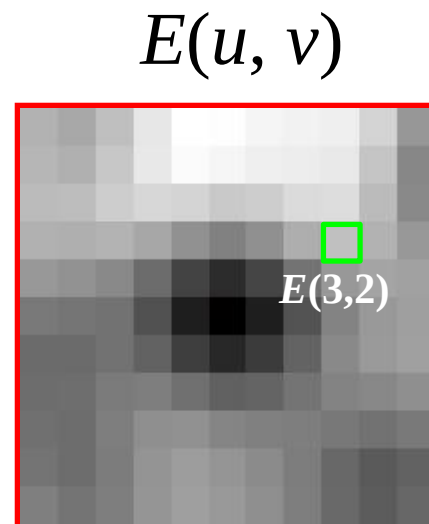
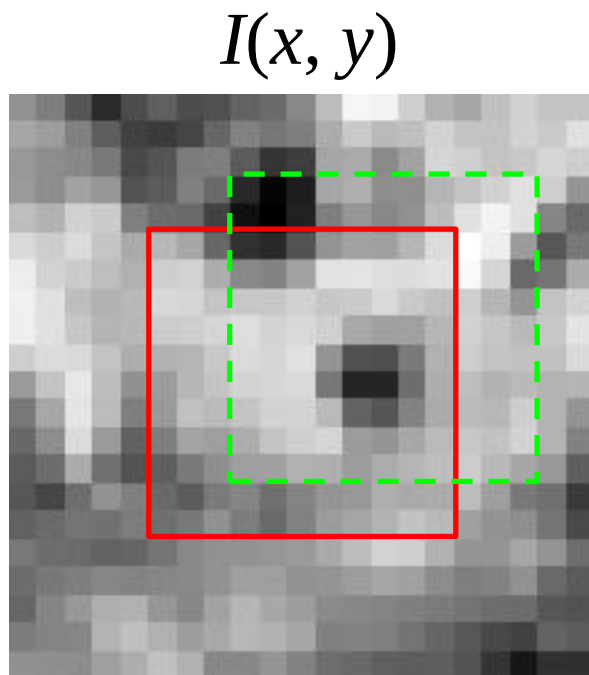


# Corner Detection: Mathematics

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Change in appearance of window  $W$  for the shift  $[u, v]$ :

$$E(u, v) = \sum_{(x,y) \in W} [I(x+u, y+v) - I(x, y)]^2$$



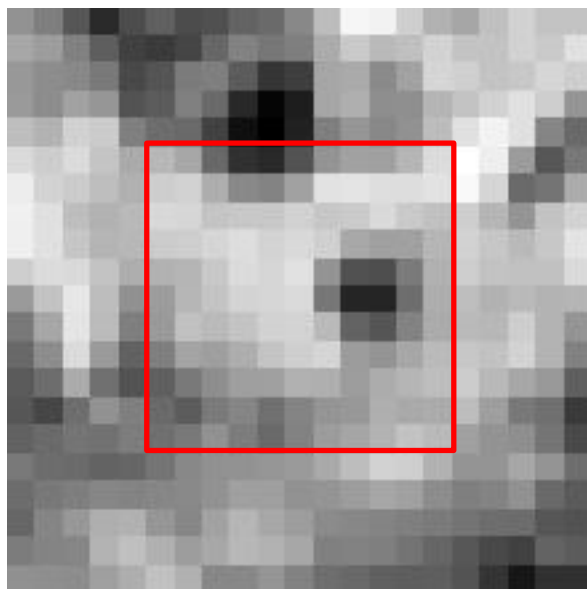
# Corner Detection: Mathematics

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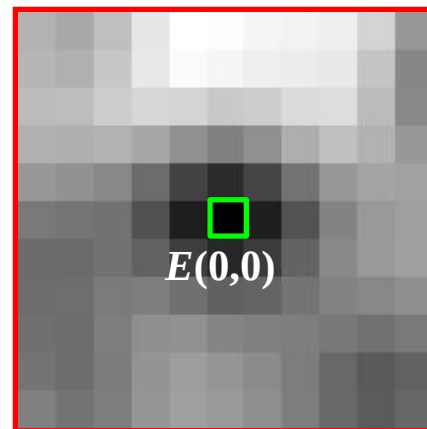
Change in appearance of window  $W$  for the shift  $[u, v]$ :

$$E(u, v) = \sum_{(x,y) \in W} [I(x+u, y+v) - I(x, y)]^2$$

$I(x, y)$



$E(u, v)$



# Corner Detection: Mathematics

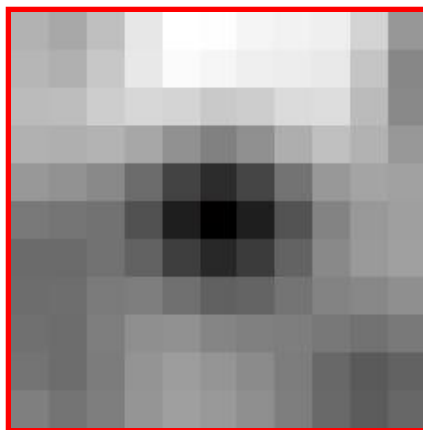
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Change in appearance of window  $W$  for the shift  $[u, v]$ :

$$E(u, v) = \sum_{(x,y) \in W} [I(x+u, y+v) - I(x, y)]^2$$

We want to find out how this function behaves for small shifts

$E(u, v)$



# Corner Detection: Mathematics

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- First-order Taylor approximation for small motions  $[u, v]$ :

$$I(x + u, y + v) \approx I(x, y) + I_x u + I_y v$$

- Let's plug this into  $E(u, v)$ :

$$E(u, v) = \sum_{(x,y) \in W} [I(x + u, y + v) - I(x, y)]^2$$



# Corner Detection: Mathematics

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The quadratic approximation can be written as

$$E(u, v) \approx u^2 \underset{x,y}{I_x^2} + 2uv \underset{x,y}{I_x I_y} + v^2 \underset{x,y}{I_y^2}$$

$$= \begin{pmatrix} u & v \end{pmatrix} \begin{pmatrix} \underset{x,y}{I_x^2} & \underset{x,y}{I_x I_y} \\ \underset{x,y}{I_x I_y} & \underset{x,y}{I_y^2} \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

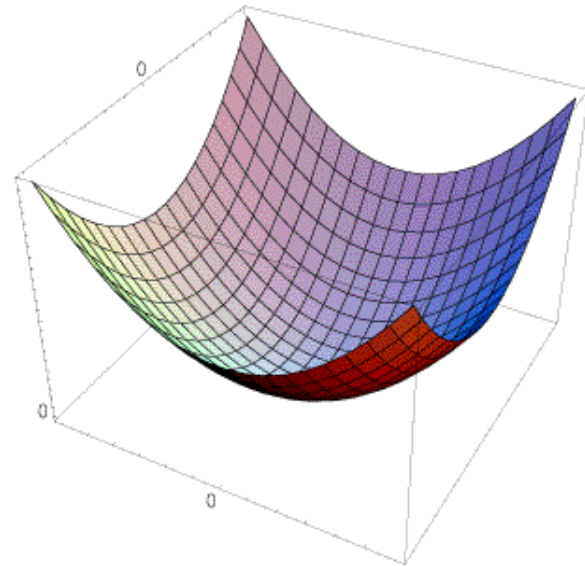
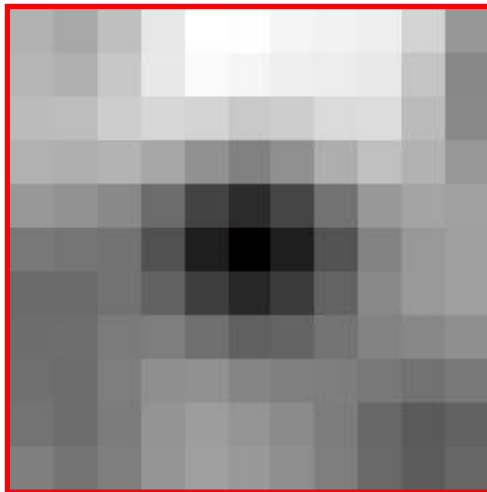
*Second moment matrix*  $M$

# Interpreting the second moment matrix

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- The surface  $E(u, v)$  is locally approximated by a quadratic form. Let's try to understand its shape.
- Specifically, in which directions does it have the fastest/slowest change?

$$E(u, v)$$

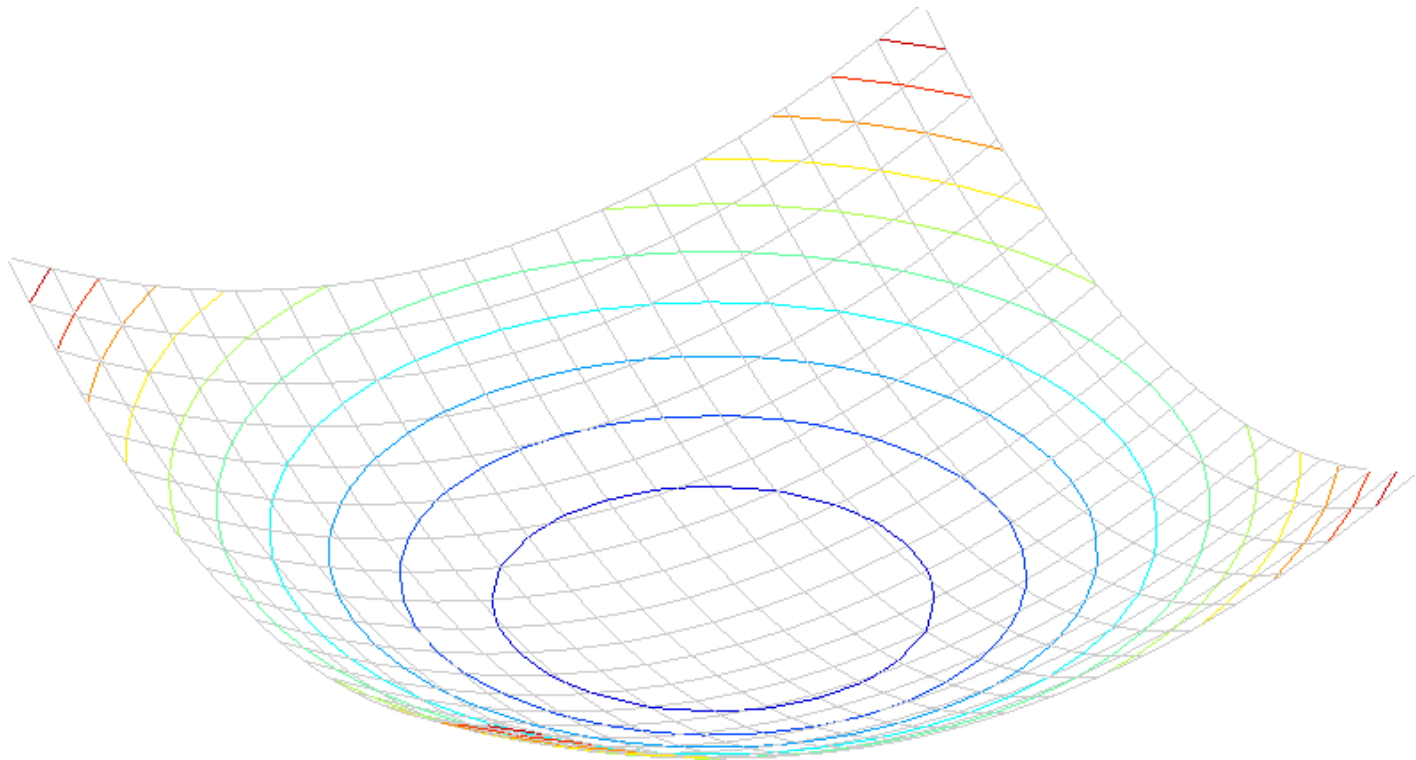


# Interpreting the second moment matrix

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Consider a horizontal “slice” of  $E(u, v)$ :  $[u \ v] M \begin{bmatrix} u \\ v \end{bmatrix} = \text{const}$

This is the equation of an ellipse:

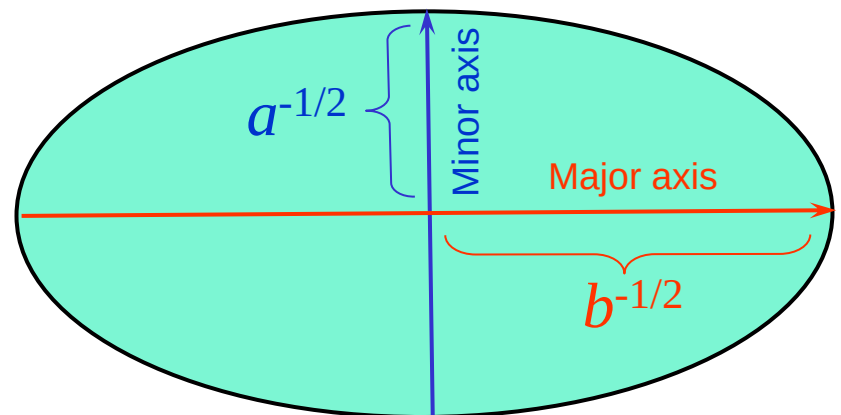


# Interpreting the second moment matrix

Consider the axis-aligned case (gradients are either horizontal or vertical)

$$M = \begin{bmatrix} \langle I_x^2 \rangle_{x,y} & \langle I_x I_y \rangle_{x,y} \\ \langle I_x I_y \rangle_{x,y} & \langle I_y^2 \rangle_{x,y} \end{bmatrix}$$

$$\begin{bmatrix} u & v \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = 1$$



# Interpreting the second moment matrix

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Consider the axis-aligned case (gradients are either horizontal or vertical)

$$M = \begin{pmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

If either  $a$  or  $b$  is close to 0, then this is **not** a corner, so we want locations where both are large

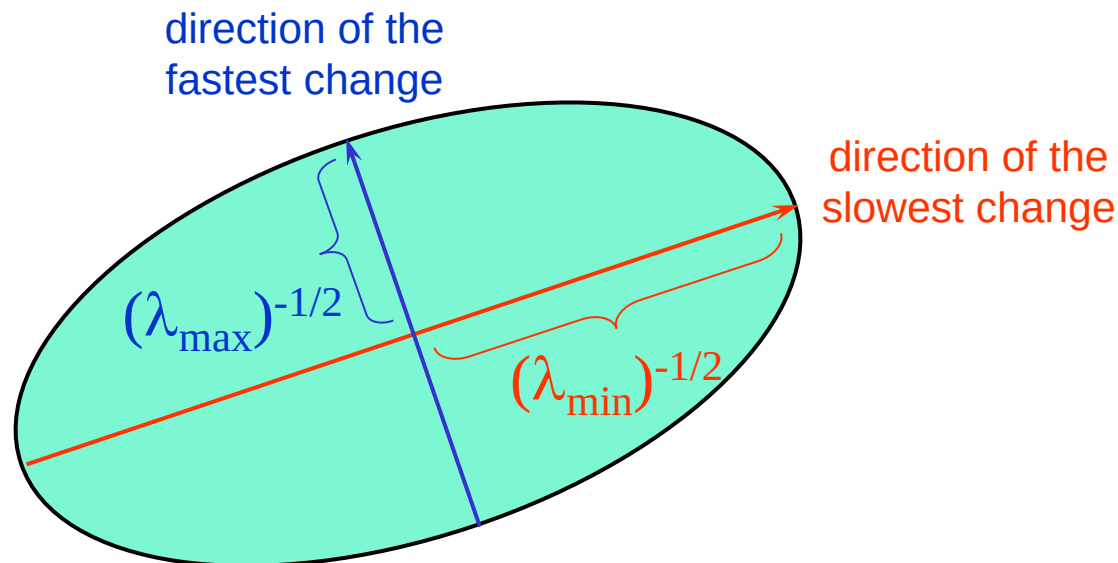
# Interpreting the second moment matrix

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In the general case, need to *diagonalize*  $M$ :

$$M = R^{-1} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} R$$

The axis lengths of the ellipse are determined by the eigenvalues and the orientation is determined by  $R$ :





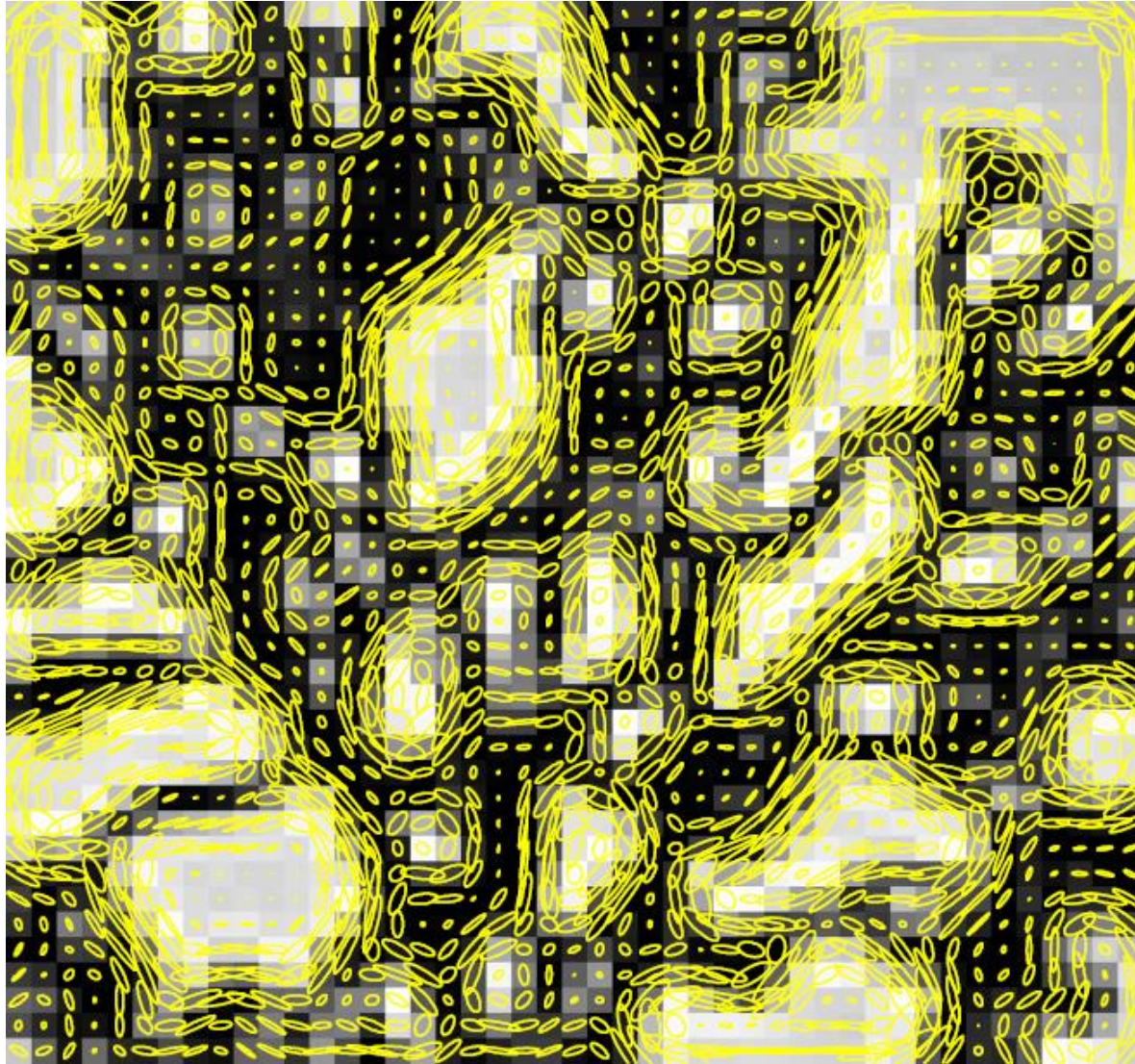
# Visualization of second moment matrices

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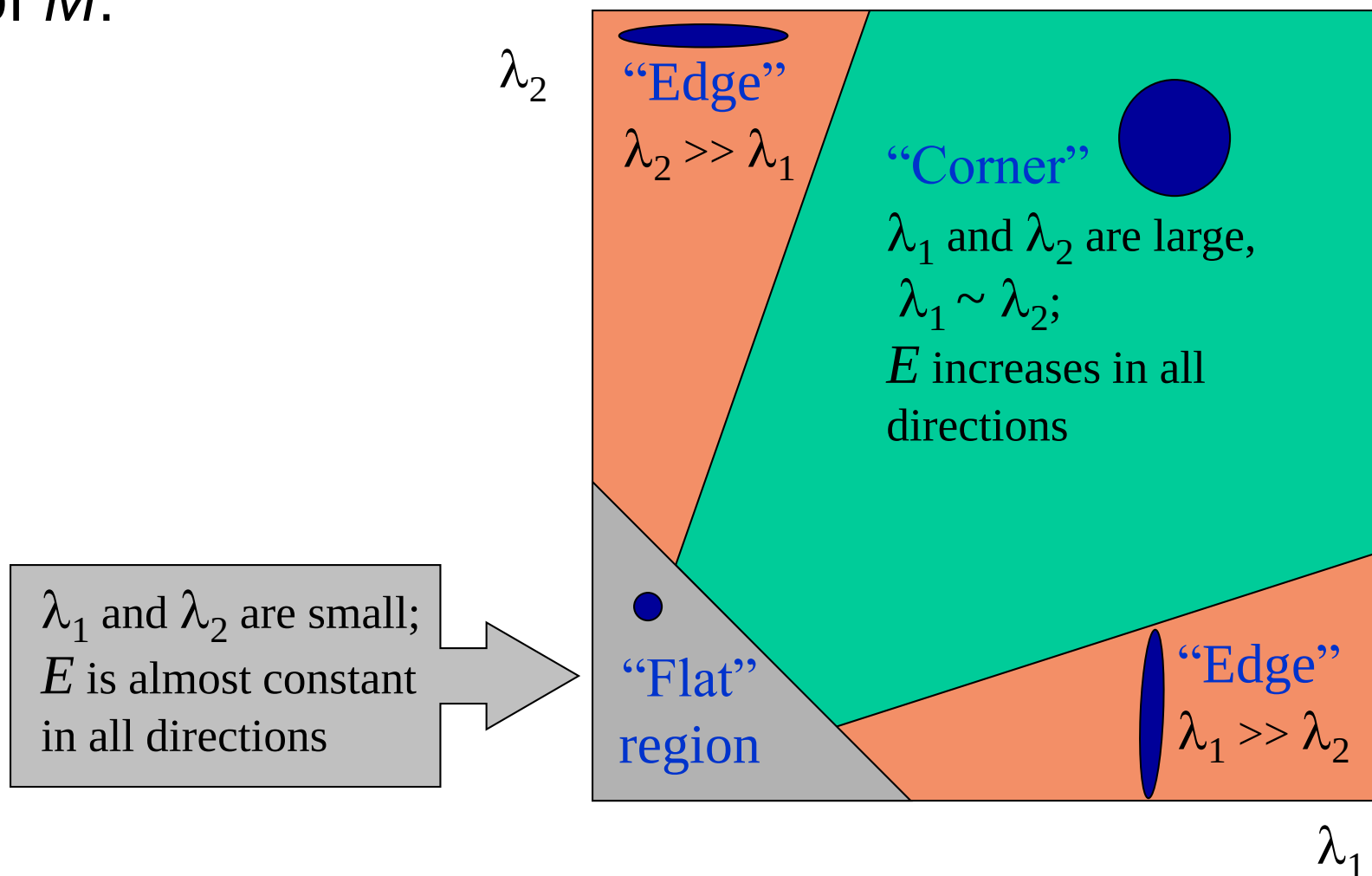
# Visualization of second moment matrices

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# Interpreting the eigenvalues

Classification of image points using eigenvalues of  $M$ :

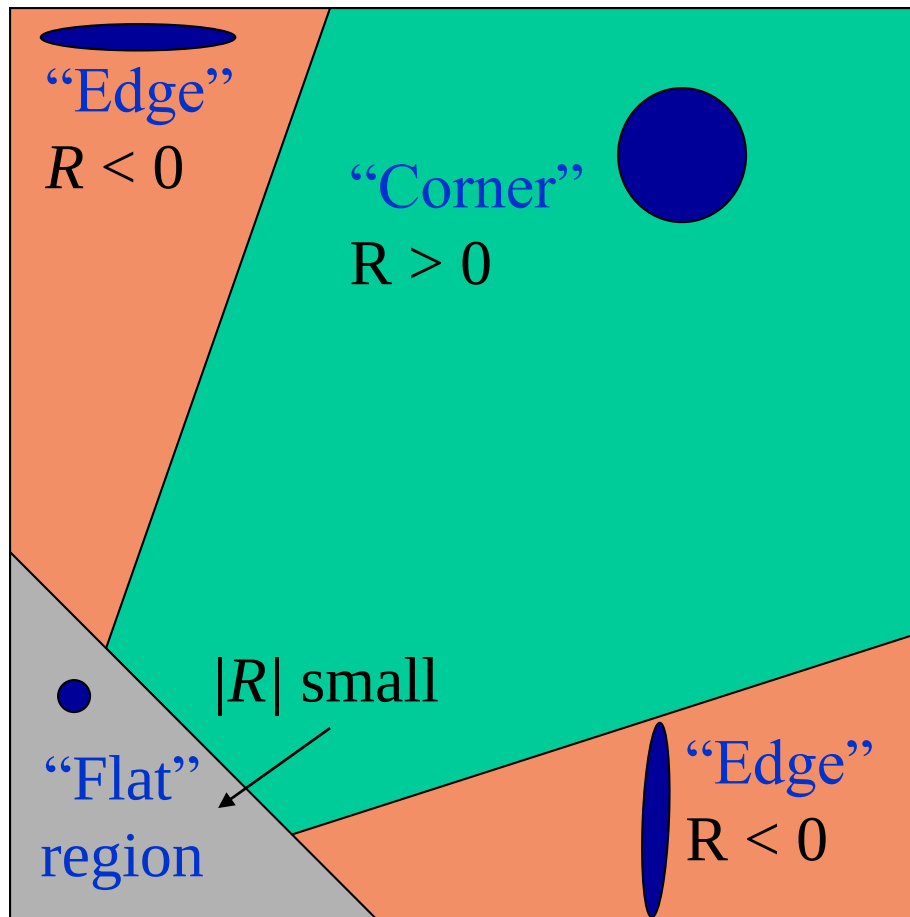


# Corner response function

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$$R = \det(M) - \alpha \text{trace}(M)^2 = \lambda_1 \lambda_2 - \alpha (\lambda_1 + \lambda_2)^2$$

$\alpha$ : constant (0.04 to 0.06)



# The Harris corner detector

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1. Compute partial derivatives at each pixel
2. Compute second moment matrix  $M$  in a Gaussian window around each pixel:

$$M = \begin{bmatrix} \sum_{x,y} w(x,y) I_x^2 & \sum_{x,y} w(x,y) I_x I_y \\ \sum_{x,y} w(x,y) I_x I_y & \sum_{x,y} w(x,y) I_y^2 \end{bmatrix}$$

C.Harris and M.Stephens, [A Combined Corner and Edge Detector](#),  
*Proceedings of the 4th Alvey Vision Conference*: pages 147—151, 1988.

# The Harris corner detector

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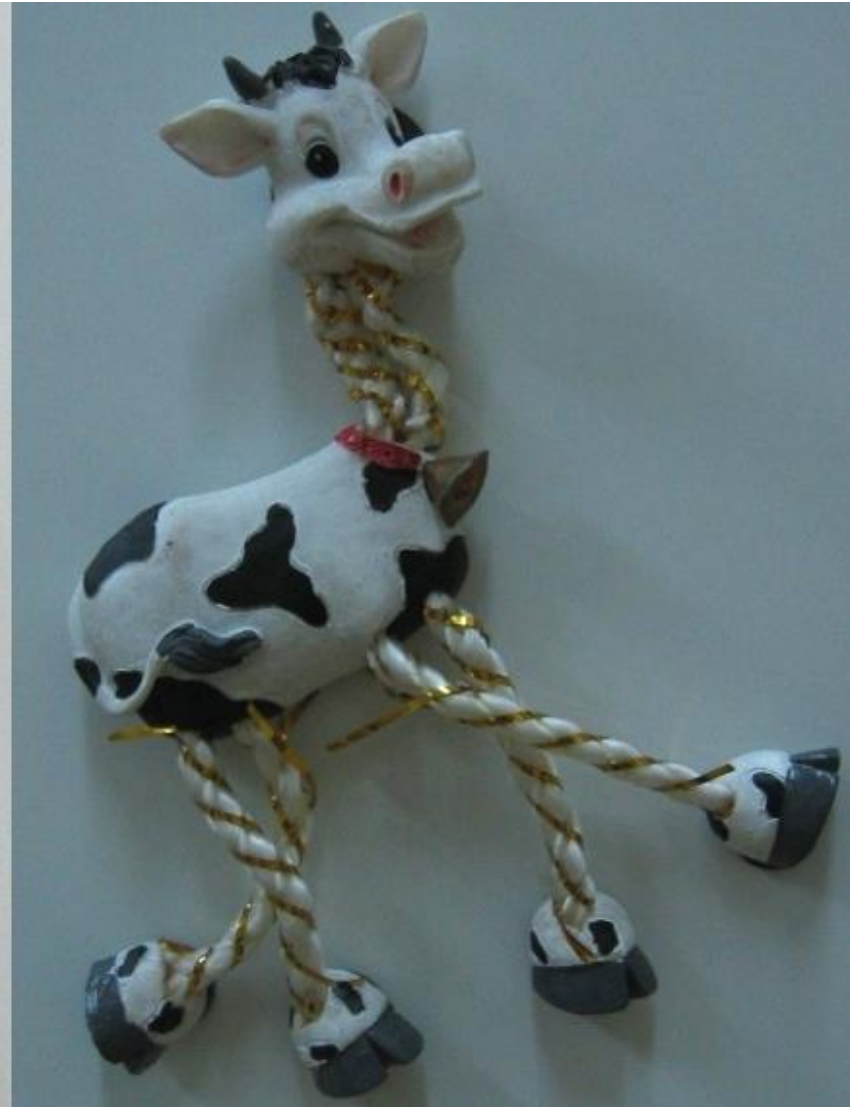
1. Compute partial derivatives at each pixel
2. Compute second moment matrix  $M$  in a Gaussian window around each pixel
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C.Harris and M.Stephens, [A Combined Corner and Edge Detector](#),  
*Proceedings of the 4th Alvey Vision Conference*: pages 147—151, 1988.



# Harris Detector: Steps

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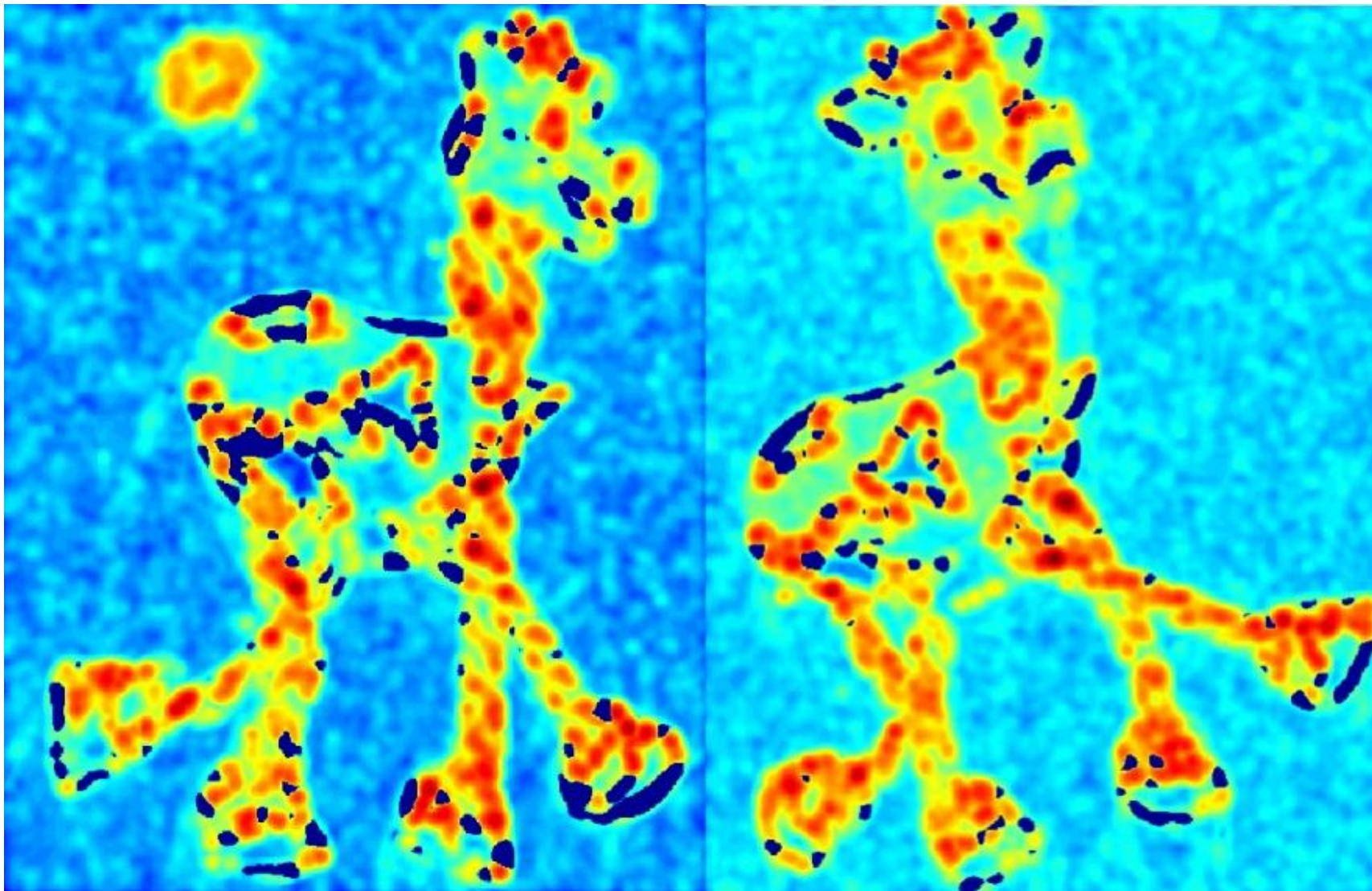




# Harris Detector: Steps

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Compute corner response  $R$



# The Harris corner detector

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1. Compute partial derivatives at each pixel
2. Compute second moment matrix  $M$  in a Gaussian window around each pixel
3. Compute corner response function  $R$
4. Threshold  $R$
5. Find local maxima of response function (nonmaximum suppression)

C.Harris and M.Stephens, [A Combined Corner and Edge Detector](#),  
*Proceedings of the 4th Alvey Vision Conference*: pages 147—151, 1988.

# Harris Detector: Steps

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Find points with large corner response:  $R > \text{threshold}$



# Harris Detector: Steps

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Take only the points of local maxima of  $R$



# Harris Detector: Steps

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# Robustness of corner features

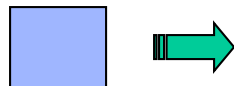
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- What happens to corner features when the image undergoes geometric or photometric transformations?



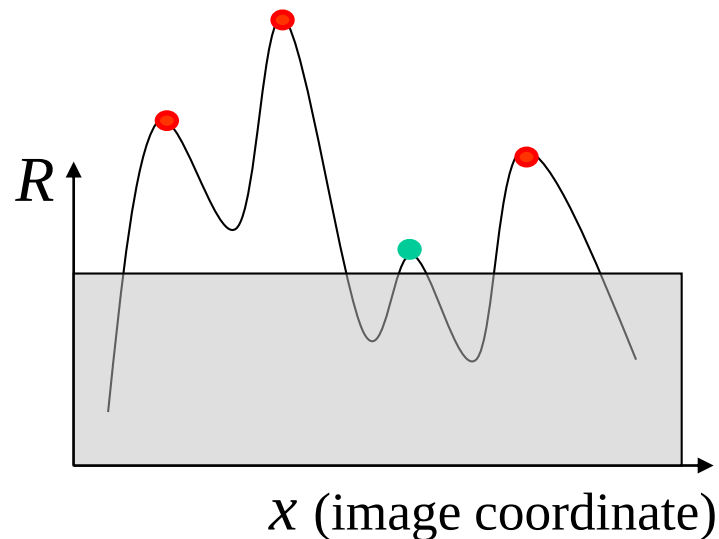
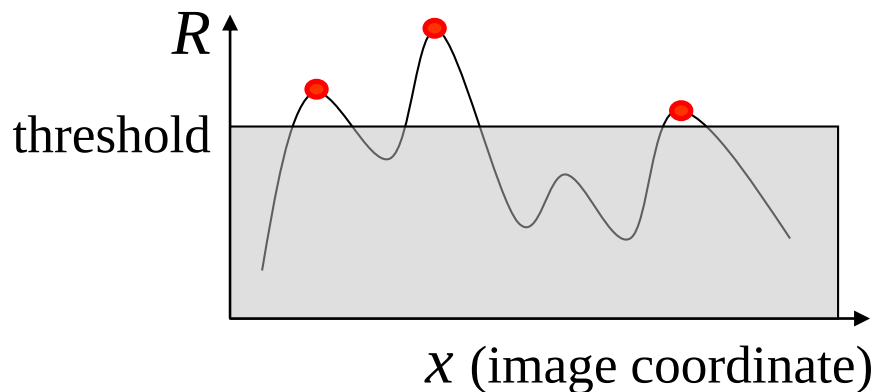
# Affine intensity change

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$$I \rightarrow a I + b$$

- Only derivatives are used => invariance to intensity shift  $I \rightarrow I + b$
- Intensity scaling:  $I \rightarrow a I$

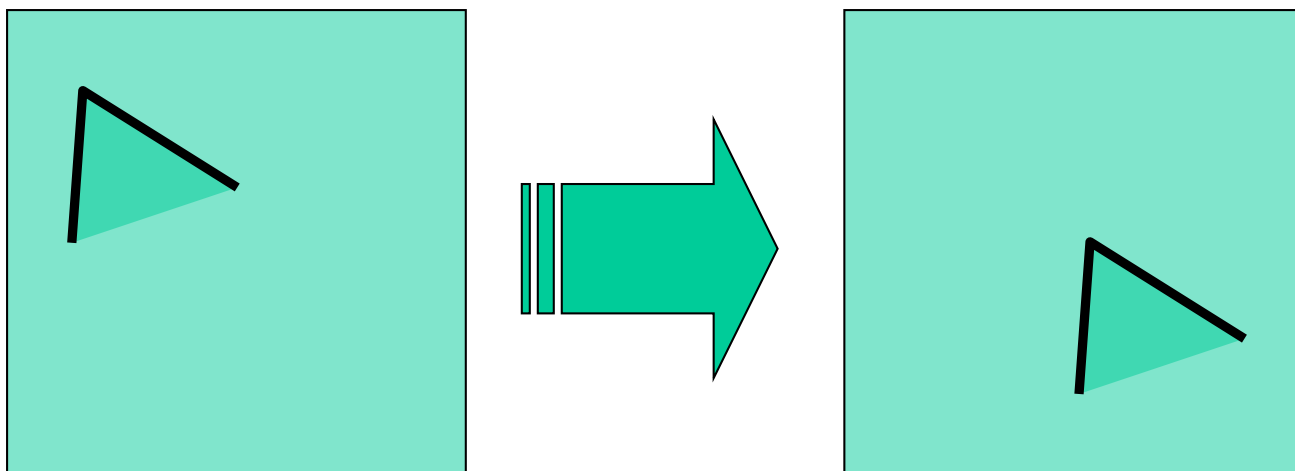


*Partially invariant to affine intensity change*



# Image translation

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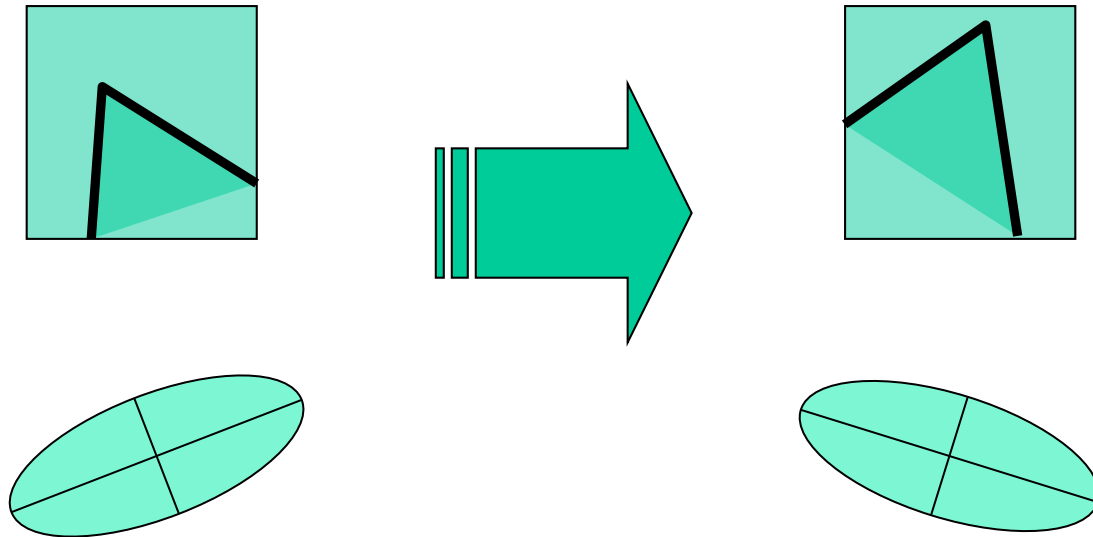


- Derivatives and window function are shift-invariant

Corner location is *covariant* w.r.t. translation

# Image rotation

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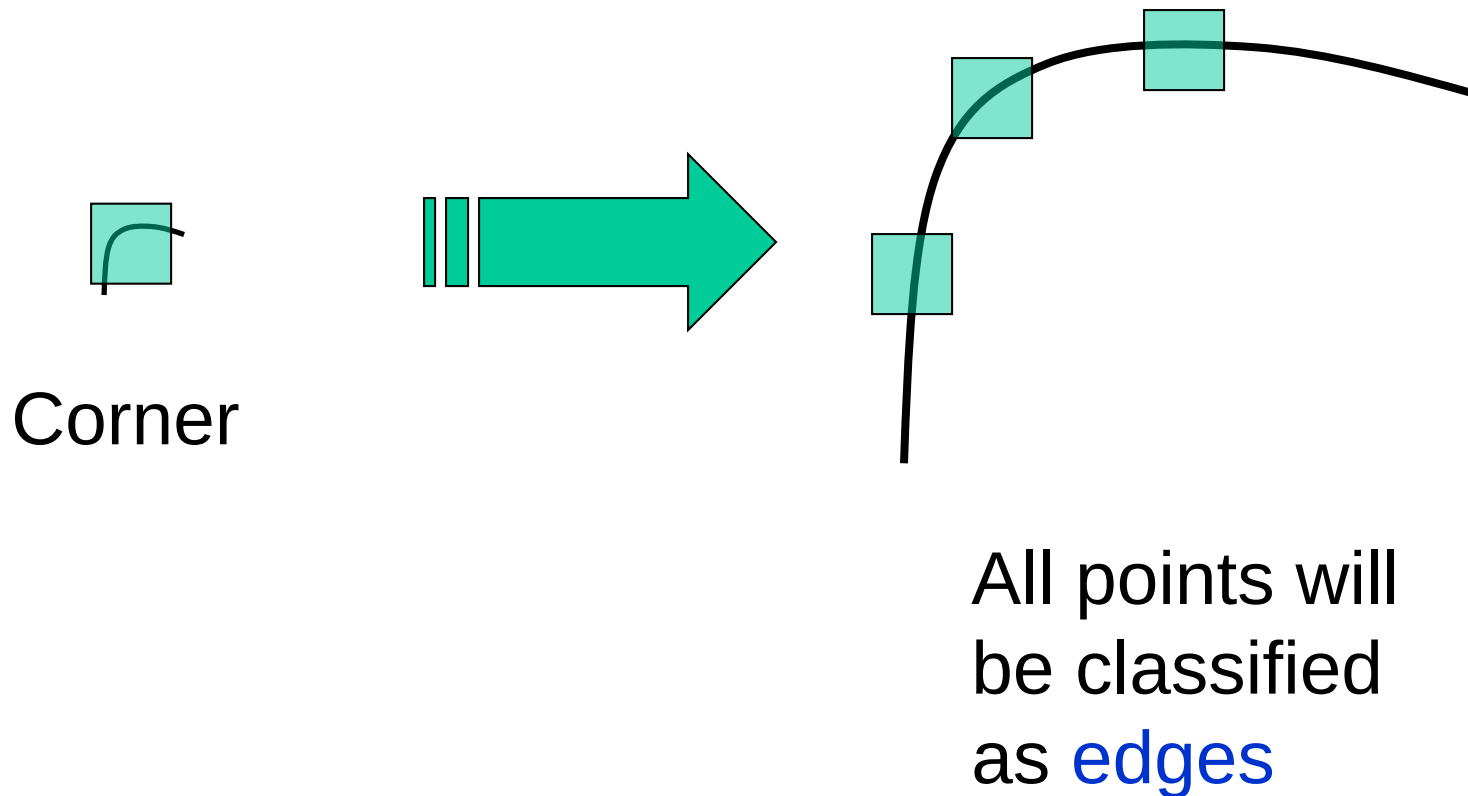


Second moment ellipse rotates but its shape (i.e. eigenvalues) remains the same

Corner location is covariant w.r.t. rotation

# Scaling

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Corner location is not covariant w.r.t. scaling!